

Credit default: rank risk, then choose an action

How well can we identify default risk?

What changes when missed defaults cost more?

Four models. One auditable comparison.

12 MIN + 3 MIN Q&A

SC6122 • Group 5

Four speakers × three minutes

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One split, shared folds, distinct responsibilities

30,000 clients • 23 predictors • 22.12% default

Categories learned within each training fold

Validation selects policies; historical test reports outcomes

18,000 → 6,000 → 6,000

FIT: five CV folds, seed 42

VALIDATION: threshold selection

TEST: fixed historical holdout

Logistic regression: a transparent reference, with a caveat

- L2 model + fold-trained scaling + one-hot categories

- 12 declared settings; CV selected
C=0.1, unweighted

- Original LR notebook was empty;
original process unverified

AP 0.4967

Baseline AP 0.4970

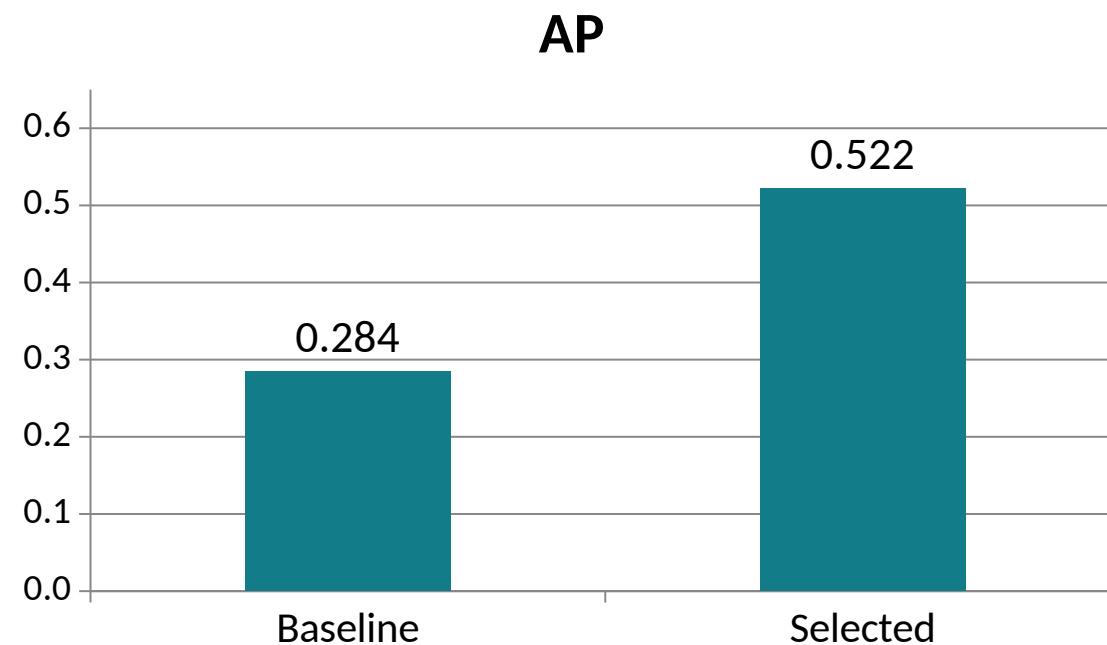
No ranking improvement from selection

LR* is a separate post-hoc supplement

Decision tree: constrain leaves to reduce overfitting

- 21 configurations: depth $\times$ minimum leaf size
- Unrestricted baseline fits training folds almost perfectly
- Selected minimum leaf size: 100 observations

Test average precision



A large baseline gap signals overfitting.

CV selection improves this tree's ranking.

Tree rules make a predictive association inspectable

- Root rule: recent repayment status
 $\text{PAY_0} \leq 1.5$

- Selected tree: depth 18, with 134 leaves

- Node proportions describe this fitting sample

16.51% | 70.13%

Fit-row default proportions

Left branch | Right branch

Association, not a causal effect

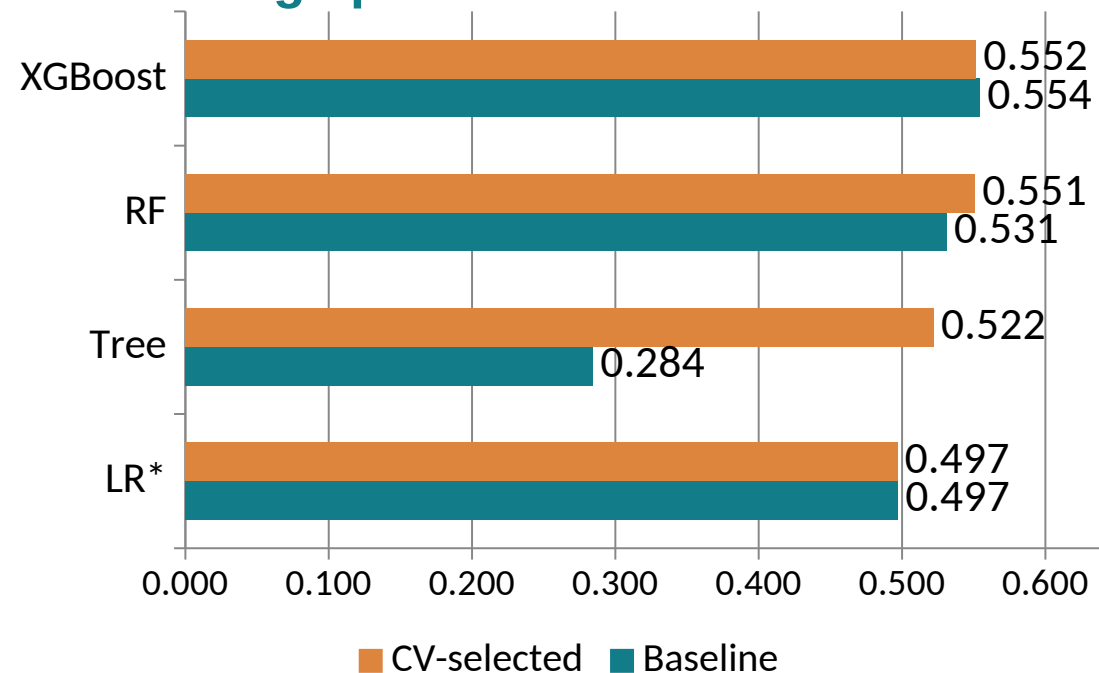
Compare ranking point estimates without declaring a winner

AP emphasizes positive-class retrieval

AP is not trapezoidal PR-AUC

CV selection precedes these test comparisons

Test average precision



Baseline and selected models are both retained.

Small ensemble differences do not establish superiority.

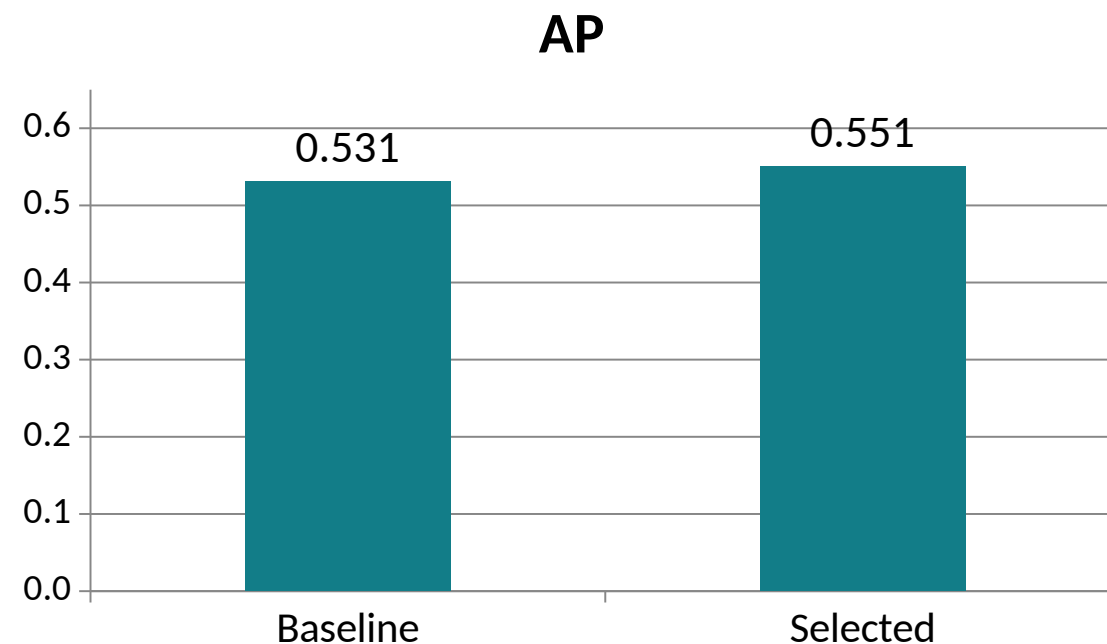
Random forest: average trees, then tune complexity

- Baseline + 23 seeded candidate settings

- Selected: 500 trees, depth 8, minimum leaf 2

- Positive-class weight 3; feature fraction 0.5

Test average precision



Averaging reduces dependence on one tree.

Weights also change the score distribution.

Forest policy: fewer misses, more review alerts

Validation selected $r=5$ threshold:
0.3229

Recall: 52.0% → 77.1%

Alert rate: 21.1% → 48.4%

3,758 → 3,404

Test cost = $FP + 5 \times FN$

Error cases checked under each policy

Same extreme cases can legitimately recur

What the audit establishes—and what it cannot

- Identical fit / validation / five-fold memberships

- Metrics recomputed after row-ID alignment

- Frozen thresholds use score $\geq$ threshold at full precision

COMPARABLE $\neq$ IDENTICAL

Shared data and objective

Different search budgets

Historical exposure and LR* caveat remain

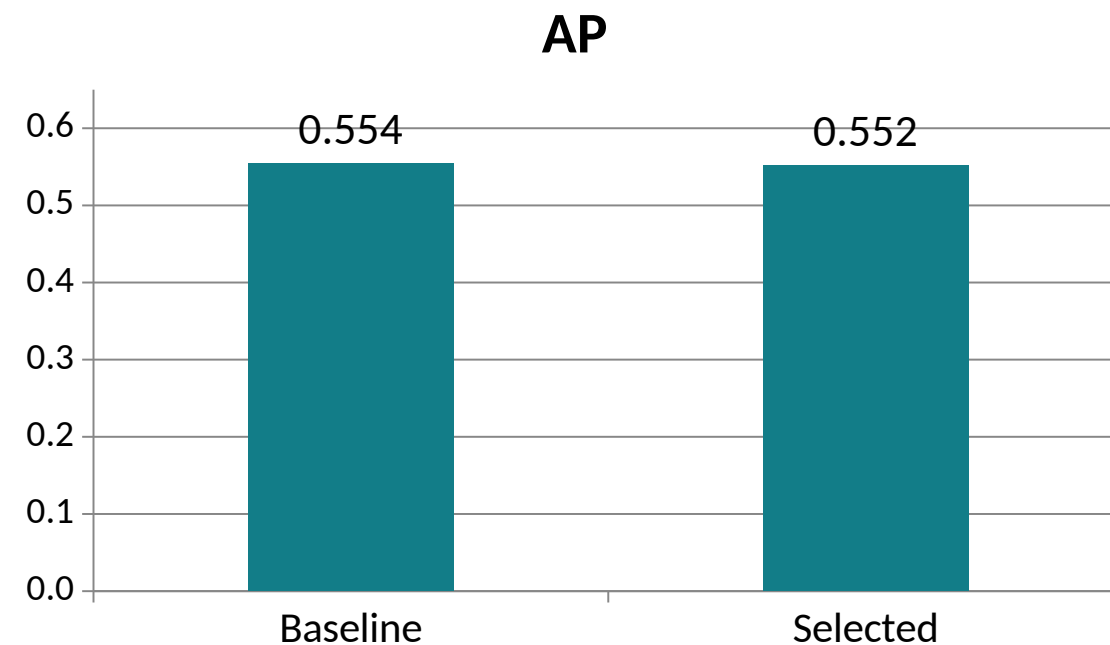
XGBoost: tuning did not improve test ranking

Baseline + 23 seeded configurations;
select by CV AP

Selected: 150 trees, depth 4, rate 0.03

Selected ROC-AUC: 0.7762

Test average precision



Tuned – baseline AP: -0.0028

Conditional 95% interval: [-0.0106, 0.0054]

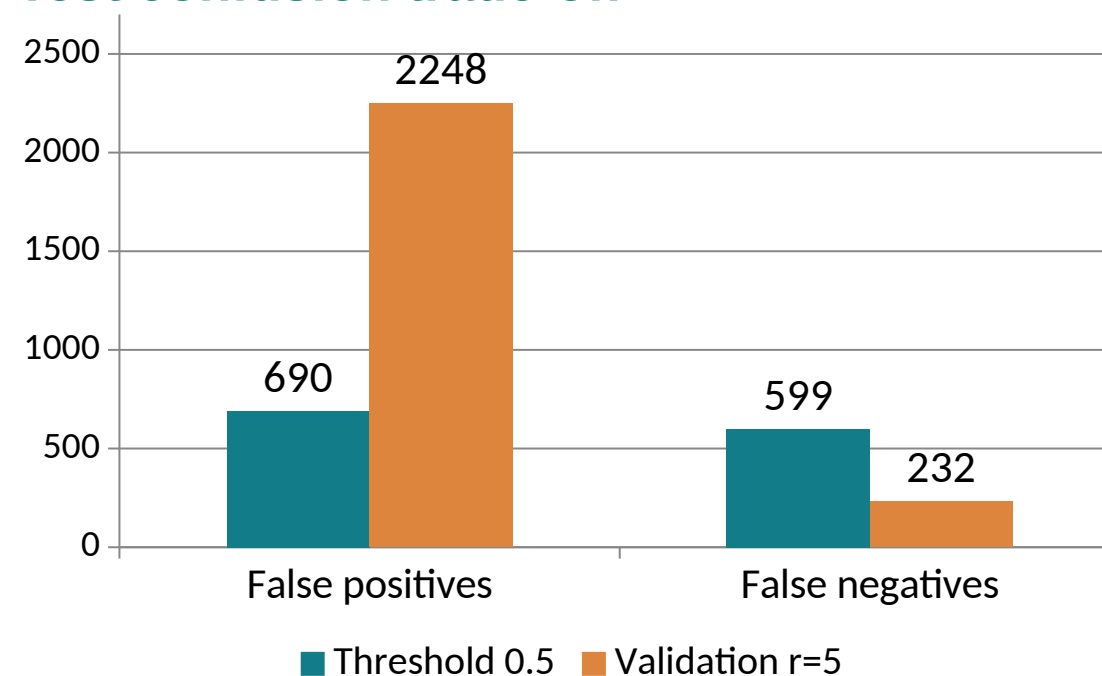
A lower threshold trades review capacity for fewer misses

Validation r=5 cutoff: 0.3150

Recall 54.9% → 82.5%; alerts 55.7%

Cost 3,685 → 3,408: 7.5% reduction

Test confusion trade-off



Same ranking scores, different action policy.

Review costs are hypothetical; capacity is unmeasured.

Choose the decision question before choosing the policy

- Ranking, explanations, and workload need separate evidence
- No clear ensemble winner from these small differences
- Next: fresh temporal data, calibration, subgroup errors

QUESTIONS • 3 MIN

Freeze realistic costs and capacity first.

Validate a new policy on new data.

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Backup A • AP, ROC-AUC and threshold metrics

AP = sum of recall increments × precision

ROC-AUC measures positive/negative score ordering

Precision, recall, F1 and accuracy require a threshold

AP ≠ TRAPEZOIDAL PR-AUC

Saved continuous scores supply ranking metrics.

Changing the cutoff changes decisions only.

Backup B • Cost ratios and fixed-prediction uncertainty

Predeclared ratios: $r = 1, 3, 5, 10$

Validation minimum cost; ties prefer largest cutoff

1,000 stratified paired bootstrap resamples

CONDITIONAL UNCERTAINTY

Fixed models / thresholds / class counts

Excludes fitting / selection uncertainty

Excludes population / prevalence shifts

Backup C • Data provenance and evidence boundaries

UCI cleaning reproduced exactly from downloaded source

EDUCATION 0/5/6 → 4; MARRIAGE 0 → 3

Retained duplicate vectors can span partitions

HISTORICAL TEST REUSED

Original LR process remains unverified.

LR* source was committed before its one run.

No new unseen-test claim is made.

Backup D • Sources, responsibilities and reproducibility

UCI: doi.org/10.24432/C55S3H

scikit-learn: average_precision_score
documentation

XGBoost paper:
doi.org/10.1145/2939672.2939785

FOUR ROLES • REVIEW REQUIRED

P1 data/LR • P2 tree/ranking

P3 forest/audit • P4 XGBoost/costs

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